

1Project Overview

Goal:
Classify 26 handwritten capital letters from 28×28 grayscale images using CNN

Application domain:
OCR · medical prescription reading
document digitisation · assistive tech

Models compared:
Custom CNN vs MobileNetV2 (Transfer Learning)

Explainability: Grad-CAM + SHAP

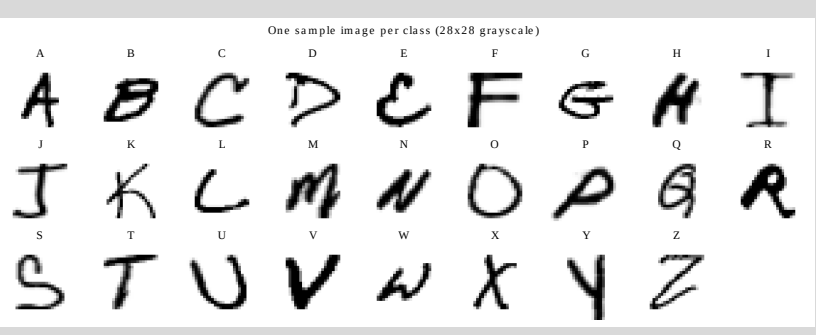
2Dataset & Split

Dataset: A–Z Handwritten Alphabets (Kaggle)
Total images: 372,451
Classes: 26 (A–Z capitals)
Image size: 28 × 28 px grayscale
Max class (O): 57,825 images
Min class (I): 1,120 images
Imbalance ratio: 51.6 : 1
Split method: Stratified sampling

70% Train

15% Val

15% Test



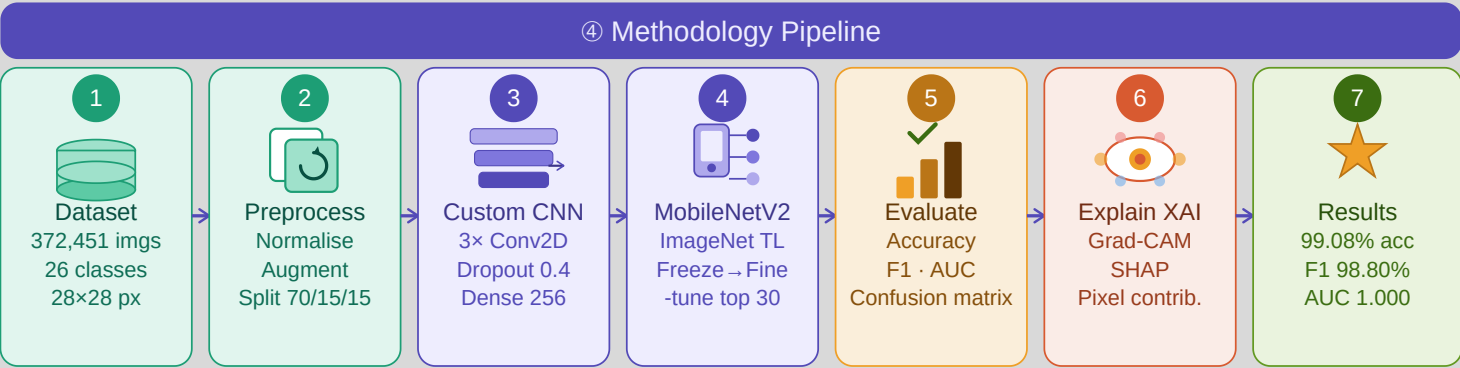
5Model Comparison (Test Set)

Model	Accuracy	Macro F1	AUC
★ Custom CNN	99.08%	98.80%	1.000
✗ MobileNetV2	19.59%	11.16%	—

Why MobileNetV2 underperformed:
Pre-trained on ImageNet (natural colour photos)
Domain mismatch with 28×28 grayscale letters

Top confused pairs (Custom CNN):
O → D (64 cases) · D → O (32 cases)
O → Q (25 cases) · W → N (13 cases)

Best model: Custom CNN
Accuracy 99.08% · F1 98.80% · AUC 1.000



6Explainable AI (XAI)

Grad-CAM
Layer: conv3 (last conv)
Heatmap on letter strokes
Letters: A E M Q R
Correct + wrong predictions

pos

neg

SHAP (DeepExplainer)
Pixel-level contributions
Red = pushes score UP
Blue = pushes score DOWN

Key finding:
Grad-CAM and SHAP both confirm model attends to stroke regions

RQ5 answered:
Model focuses on meaningful letter strokes, not background pixels

7Research Questions & Answers

RQ	Question	Answer
RQ1	Can a compact CNN achieve high accuracy?	✓ Yes
RQ2	Which letter pairs are most confused?	O ↔ D, Q
RQ3	Does imbalance reduce recall on rare letters?	✓ Fixed
RQ4	Does MobileNetV2 outperform custom CNN?	✗ No
RQ5	What does the model focus on?	Strokes ✓

Best model: Custom CNN · 99.08% · F1 98.80% · AUC 1.000

8Key Results Summary

★ 99.08%
Test accuracy (CNN)

✓ 98.80%
Macro F1
1.000 AUC

Minority class recall (class weighting effective):
F (1,163 samples) → recall = 0.994
I (1,120 samples) → recall = 0.976

Confusion matrix highlights:
Near-perfect diagonal across all 26 classes
Worst: O → D (64 of 8,674 O-class samples)